5.12 (a) Since $\pi(x,y)=x+y$ is $\langle\mathcal{B}(\mathbb{R})\times\mathcal{B}(\mathbb{R}),\mathcal{B}(\mathbb{R})\rangle$-measurable, $f(x+y)=(f\circ\pi)(x,y)$ is $\langle\mathcal{B}(\mathbb{R})\times\mathcal{B}(\mathbb{R}),\mathcal{B}(\mathbb{R})\rangle$-measurable. Clearly, $\nu=(\mu\times\lambda)\circ\pi^{-1}$. By the change of variables formula (Problem 2.42), $\int f d\nu=\int f\circ\pi d(\mu\times\lambda)$, i.e. $\int f d\nu=\int f(x+y)d\mu d\lambda$.

(b) For any $A\in\mathcal{B}(\mathbb{R})$, $\nu(A)=\int\left(\int I_{A-y}(x)d\mu(x)\right)d\lambda(y)=\int\mu(A-y)d\lambda(y)$.

(c) A natural choice of B_ν would be $B_\nu=B_\mu+B_\lambda=\{x+y\mid x\in B_\mu, y\in B_\lambda\}$. The problem will be finished if we realize that

$$I_{B_\nu^c}(x+y)\le I_{B_\mu^c}(x)+I_{B_\lambda^c}(y),\quad \forall x,y\in\mathbb{R},\quad (*)$$

because, in this way, $\nu(B_\nu^c)=\int\int I_{B_\nu^c}(x+y)d\mu d\lambda\le\int I_{B_\mu^c}(x)d\mu+\int I_{B_\lambda^c}(y)d\lambda=0$. $(*)$ is true since $x+y\in B_\nu^c\Rightarrow x+y\notin B_\mu+B_\lambda\Rightarrow x\notin B_\mu\,or\,y\notin B_\lambda\Rightarrow x\in B_\mu^c\,or\,y\in B_\lambda^c$. Basically, this problem says that sum of two independent discrete random variables is still a discrete random variable.

(d) It follows immediately from the result in (b). It's saying that the sum of two independent random variables will not be discrete as long as one of the summed two random variables is not discrete.

(e) Given that we are allowed to use whatever results we have proved in former chapters $\ddot{\smile}$, for any $A\in\mathcal{B}(\mathbb{R})$, $\nu(A)=\int\{\int I_A(x+y)h(x)dm(x)\}d\lambda(y)=\int\{\int I_A(t)h(t-y)dm(t)\}d\lambda(y)=\int I_A(t)\{\int h(t-y)d\lambda(y)\}dm(t)=\int_A\{\int h(t-y)d\lambda(y)\}dm(t)$. So $\nu\ll m$ and $\dfrac{d\nu}{dm}(t)=\int h(t-y)d\lambda(y)=h*\lambda\ a.e.(m)$.

(f) For any $A\in(\mathbb{R})$, $\nu(A)=\int\{\int I_A(x+y)\dfrac{d\mu}{dm}(x)dm(x)\}\dfrac{d\lambda}{dm}(y)dm(y)=\int\{\int I_A(t)\dfrac{d\mu}{dm}(t-y)dm(t)\}\dfrac{d\lambda}{dm}(y)dm(y)=\int_A\{\int\dfrac{d\mu}{dm}(t-y)\dfrac{d\lambda}{dm}(y)dm(y)\}dm(t)=\int_A(\dfrac{d\mu}{dm}*\dfrac{d\lambda}{dm})(t)dm(t)$. So $\nu\ll m$ and $\dfrac{d\nu}{dm}=\dfrac{d\mu}{dm}*\dfrac{d\lambda}{dm}$, $a.e.(m)$.

5.14 (a) Suppose $|f|\le M$ for some finite positive constant M, then $|f*g|\le M\int|g(x-u)|du=M\|g\|_1<\infty$. For any sequence $\{x_n\}_{n\ge1}$ whose limit is x, by DCT and the continuity of f, $\lim_{n\to\infty}(f*g)(x_n)=f*g(x)$, that is, $f*g$ is continuous.

(b) Let $h_t(x)=\{(f*g)(x+t)-(f*g)(x)\}/t$, that is, $h_t(x)=\int[\{f(x-u+t)-f(x-u)\}/t]g(u)du$. Because f is differentiable and f' is bounded, applying the continuous version of DCT, $\lim_{t\to0}h_t(x)=\int f'(x-u)g(u)du$, i.e. $(f*g)'=\int f'(x-u)g(u)du$. It is easy to see that $(f*g)'$ is bounded due to boundedness of f'.

5.15 (a) The first inequality is trivial when $p=1$. For $p>1$, by Hőlder's Inequality, $\|f(u)g(x-u)\|_1\le\||f(u)|^{1-1/p}|f(u)|^{1/p}|g(x-u)|\|_1\le\||f(u)|^{1-1/p}\|_{p/(p-1)}\cdot\||f(u)|^{1/p}|g(x-u)|\|_p$. That is

$$\int|f(u)g(x-u)|du\le\left(\int|f|dm\right)^{1-1/p}\left(\int|g(x-u)|^p|f(u)|du\right)^{1/p}.$$

The fact $\int|g(x-u)|^p|f(u)|du<\infty\,(a.e.)$ w.r.t. $m(x)$ follows from Corollary 5.4.3. As one can see, it's not necessary to prove the first inequality in order to guarantee that

$f * g$ is well-defined. Because Corollary 5.4.3 together with the fact that $g \in L^p(\mathbb{R}), p \geq 1 \Rightarrow g \in L(\mathbb{R})$ easily finish this job.

However, it is useful for proving the second inequality if one does not like Jensen's Inequality. By the first inequality,

$$\|f * g\|_p^p \leq \|f\|_1^{p-1} \int \left(\int |g(x-u)|^p |f(u)| du \right) dx = \|f\|_1^p \|g\|_p^p.$$

Notice that the above derivation uses Tonelli's Theorem and the invariance of translation property of Lebesgue measure. It's not hard to see that, by Hőlder's Inequality, the equality holds if and only if $f = 0$ or $g = 0$ $a.e.(m)$.

An alternative approach is following the hint and using Jensen's Inequality by claiming that

$$\left(\int |g(x-u)| d\mu(u) \right)^p \leq \int |g(x-u)|^p d\mu(u),$$

where $d\mu$ is the one defined in the hint. The result will then follows immediately after applying Tonelli's Theorem.

A. Since $\{f \in A\} \cap \{g \in B\} = \{(x,y) : x \in A\} \cap \{(x,y) : y \in B\} = \{(x,y) : x \in A, y \in B\} = \{(x,y) : (x,y) \in A \times B\} = \{(f,g) \in A \times B\}$, then $\nu(\{f \in A\} \cap \{g \in B\}) = \nu(\{(f,g) \in A \times B\})$. By Theorem 5.1.2 (iii), we have $\nu(\{(f,g) \in A \times B\}) = \mu(\{f \in A\})\lambda(\{g \in B\})$. Clearly $\nu(\{f \in A\}) = \mu(\{f \in A\})$ and $\nu(\{g \in A\}) = \lambda(\{g \in A\})$ since μ, λ are probability measures. So $\nu(\{(f,g) \in A \times B\}) = \nu(\{f \in A\})\nu(\{g \in B\})$.

Since $f,\ g \in L^2(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2), \nu)$, then by Cauchy Schwarz inequality, we have $|\int fg\ d\nu| \leq (\int f^2 d\nu)^{\frac{1}{2}} (\int g^2 d\nu)^{\frac{1}{2}} < \infty$. Similarly $|\int f d\nu| < \infty$ and $|\int g d\nu| < \infty$. Then

$$\begin{aligned} \int fg\ d\nu &= \int xy\ d\mu \times \lambda \\ &= \int x d\mu \int y d\lambda \quad \text{(since } \mu, \lambda \text{ are probability measures)} \\ &= \int f d\nu \int g d\nu \end{aligned}$$

Interpretation: random variables X and Y are independent, i.e. $P_\nu(X \in A,\ Y \in B) = P_\nu(X \in A) P_\nu(Y \in B)$, then $EXY = EXEY$.

6.1 Two such measures are the product measure $\mu_1 \times \mu_2$ and λ, where $\lambda((\omega_1, \omega_2)) = 1/4$ for all $(\omega_1, \omega_2) \in \Omega \times \Omega$.

6.3 (a) In order to specify the distributions P_X and P_Y on $(\mathbb{R}, (\mathbb{R}))$, it suffices to give the CDFs of them. Let Φ be the CDF of $N(0,1)$, then $P_X((-\infty, x]) = P(Z^2 \leq x) = \{\Phi(\sqrt{x}) - \Phi(-\sqrt{x})\} I_{[0,\infty)}(x)$, $P_Y((-\infty, y]) = P(e^{-X} \leq y) = P(X \geq -\log y) I_{(0,\infty)}(y) =$

$I_{[1,\infty)}(y) + 2\Phi(-\sqrt{-\log y})I_{(0,1)}(y)$. Now let us calculate the three integrals:

$$\int_{\mathbb{R}} e^{-z^2}\phi(z)dz = \frac{1}{\sqrt{3}}\int_{\mathbb{R}} \phi(\sqrt{3}z)d(\sqrt{3}z) = 3^{-1/2}.$$

$$\begin{aligned}\int_{\mathbb{R}} e^{-x}dP_X(x) &= \int_{\mathbb{R}^+} e^{-x}\phi(\sqrt{x})/\sqrt{x}dx = 2\int_{\mathbb{R}^+} e^{-(\sqrt{x})^2}\phi(\sqrt{x})d\sqrt{x} \\ &= 2\int_{\mathbb{R}^+} e^{-z^2}\phi(z)dz = 3^{-1/2}.\end{aligned}$$

$$\begin{aligned}\int_{\mathbb{R}} ydP_Y(y) &= \int_0^1 \phi(-\sqrt{-\log y})/(\sqrt{-\log y})dy = \int_0^1 2y\phi(-\sqrt{-\log y})d(-\sqrt{-\log y}) \\ &= \int_0^1 2e^{(-\sqrt{-\log y})^2}\phi(-\sqrt{-\log y})d(-\sqrt{-\log y}) \\ &= 2\int_{-\infty}^0 e^{-z^2}\phi(z)dz = 3^{-1/2}.\end{aligned}$$

(b) $Y \sim N(0,k)$, $Z/k \sim \chi_1^2$. To be strict, one can verify this by getting the convolution of sum of two independent standard normally distributed random variables first, then apply it to the sum up to k.
Since $X_1, \ldots, X_k$ are iid, their joint probability measure are the product measure $\times_{i=1}^k P_{X_i} = P_X^k$. As we all know the first moment and second moment of $N(0,10$ are 0 and 1 respectively, applying Tonelli's theorem, one can break the first integral down step by step. For example,

$$\begin{aligned}&\int_{\mathbb{R}^k} (x_1 + \ldots + x_k)^2 dP_{X_1,\ldots,X_k}(x_1,\ldots,x_k) \\ &= \int_{\mathbb{R}^{k-1}} \left\{\int_{\mathbb{R}} [x_1^2 + 2x_1(x_2+\ldots+x_k) + (x_2+\ldots+x_k)^2]dP_X(x_1)\right\} dP_X^{k-1}(x_2,\ldots,x_k) \\ &= 1 + \int_{\mathbb{R}^{k-1}} (x_2+\ldots+x_k)^2 dP_X^{k-1}(x_2,\ldots,x_k) = \cdots = k.\end{aligned}$$

Since we already know that $Y \sim N(0,k)$, $\int_{\mathbb{R}} y^2 dP_Y(y) = k$. Moreover, $\int_{\mathbb{R}} zdP_Z(z) = k$ as the expectation of $k\chi_1^2$.

(c) Either by applying convolution or using the practical meaning of binomial distribution, one can easily verify that $Y \sim \text{Binomial}(\sum_{i=1}^k n_i, p)$. Use the same argument again, $\int_{\mathbb{R}^k}(x_1 + \ldots + x_k)dP_{X_1,\ldots,X_k}(x_1,\ldots,x_k) = \sum_{i=1}^k \int_{\mathbb{R}} x_i dP_{X_i}(x_i) = (\sum_{i=1}^k n_i)p$. Clearly, $\int_{\mathbb{R}} ydP_Y(y) = (\sum_{i=1}^k n_i)p$ using the mean formula of a binomial distribution.

6.6 It is trivial when either of X and Y is a constant. Otherwise, let $a = Cov(X,Y)/Var(X), b = EY - aEX$. When equality of (2.20) holds, $Var(Y - aX) = VarY - Cov(X,Y)^2/Var(X) = 0$, that is, $Y - aX = EY - aEX$ $a.s.$